

Algoritmo del simplesso duale

Exercise

the buyer's problem

$$\begin{aligned}\min \varphi &= 4x_1 + 5x_2 \\ \text{s.a} \quad x_1 + \frac{1}{2}x_2 &\geq 1 \\ x_2 &\geq 1 \\ x_1, x_2 &\geq 0\end{aligned}$$

the producer's problem

$$\begin{aligned}\max \omega_D &= y_1 + y_2 \\ \text{s.a} \quad y_1 &\leq 4 \\ \frac{1}{2}y_1 + y_2 &\leq 5 \\ y_1, y_2 &\geq 0\end{aligned}$$

Exercise

the buyer's problem

$$\begin{aligned}\min \varphi &= 4x_1 + 5x_2 \\ \text{s.a} \quad x_1 + \frac{1}{2}x_2 &\geq 1 \\ x_2 &\geq 1 \\ x_1, x_2 &\geq 0\end{aligned}$$

standard form

$$\begin{aligned}\min \varphi &= 4x_1 + 5x_2 \\ \text{s.a} \quad x_1 + \frac{1}{2}x_2 - s_1 &= 1 \\ x_2 - s_2 &= 1 \\ x_1, x_2, s_1, s_2 &\geq 0\end{aligned}$$

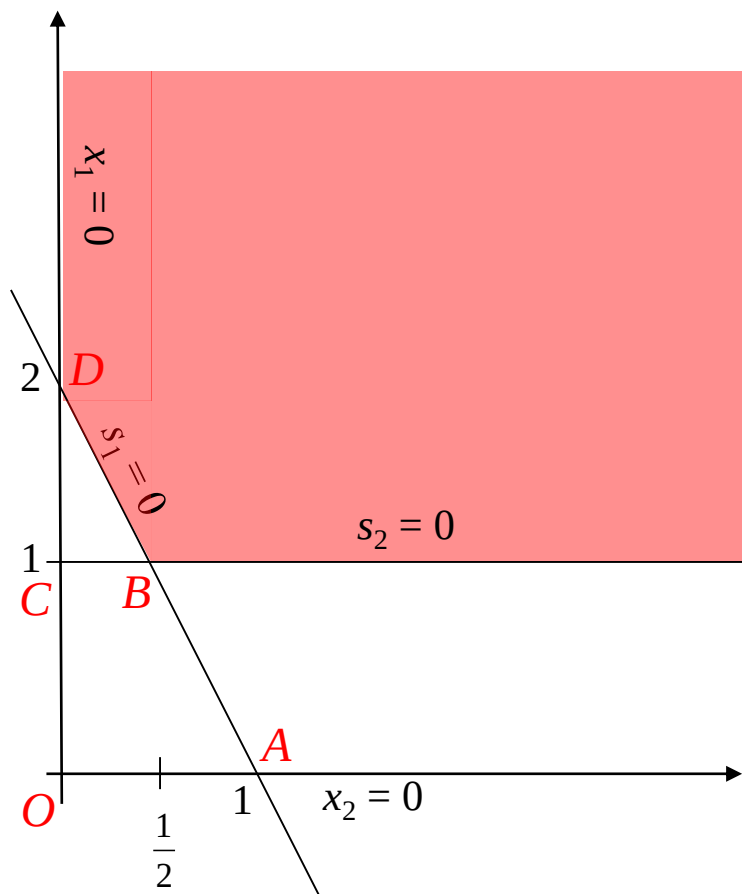
the producer's problem

$$\begin{aligned}\max \omega_D &= y_1 + y_2 \\ \text{s.a} \quad y_1 &\leq 4 \\ \frac{1}{2}y_1 + y_2 &\leq 5 \\ y_1, y_2 &\geq 0\end{aligned}$$

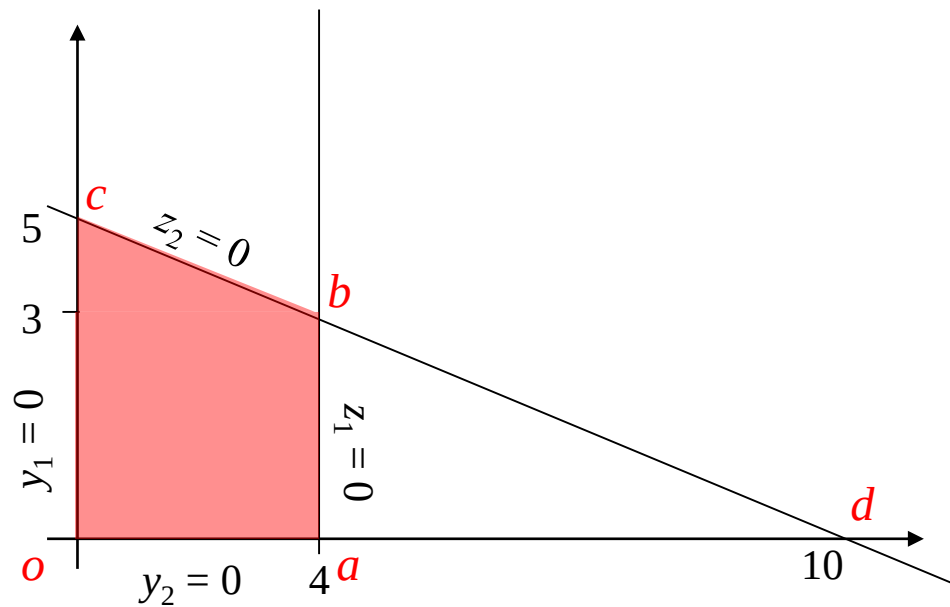
standard form

$$\begin{aligned}\min \varphi_D &= -y_1 - y_2 \\ \text{s.a} \quad y_1 + z_1 &= 4 \\ \frac{1}{2}y_1 + y_2 + z_2 &= 5 \\ z_1, z_2, y_1, y_2 &\geq 0\end{aligned}$$

the buyer's problem



the producer's problem



Vertex O :

$$\begin{array}{ccccc|c}
 x_1 & x_2 & s_1 & s_2 & \varphi & \\
 \hline
 -1 & -\frac{1}{2} & 1 & 0 & 0 & -1 \\
 0 & -1 & 0 & 1 & 0 & -1 \\
 \hline
 -4 & -5 & 0 & 0 & 1 & 0
 \end{array}$$

Basic solution

- Infeasible
- **Optimal**

Ricaviamo il vertice nel problema duale:

$$\begin{array}{c|cc|c}
 & x_1 & x_2 & \\
 \hline
 s_1 & -1 & -\frac{1}{2} & -1 \\
 s_2 & 0 & -1 & -1 \\
 \hline
 \varphi & -4 & -5 & 0
 \end{array}$$

$$\begin{array}{c|cc|c}
 & y_1 & y_2 & \\
 \hline
 z_1 & 1 & 0 & 4 \\
 z_2 & \frac{1}{2} & 1 & 5 \\
 \hline
 \varphi_D & 1 & 1 & 0
 \end{array}$$

$$\begin{array}{ccccc|c}
 z_1 & z_2 & y_1 & y_2 & \varphi_D & \\
 \hline
 1 & 0 & 1 & 0 & 0 & 4 \\
 0 & 1 & \frac{1}{2} & 1 & 0 & 5 \\
 \hline
 0 & 0 & 1 & 1 & 1 & 0
 \end{array}$$

vertice o: s.b.a. non ottima

vertice o

$$\begin{array}{c}
 \begin{array}{ccccc}
 z_1 & z_2 & y_1 & y_2 & \varphi_D \\
 \hline
 1 & 0 & 1 & 0 & 0 \\
 0 & 1 & \frac{1}{2} & 1 & 0 \\
 0 & 0 & 1 & 1 & 1
 \end{array}
 \left| \begin{array}{c} 4 \\ 5 \\ 0 \end{array} \right.
 \end{array}$$

$$\frac{4}{1} = 4 \quad \leftarrow \text{min}$$

$$\frac{5}{\frac{1}{2}} = 10$$

entra in base y_1 ed esce di base z_1

vertice O

$$\begin{array}{ccccc}
 x_1 & x_2 & s_1 & s_2 & \varphi \\
 \hline
 -1 & -\frac{1}{2} & 1 & 0 & 0 \\
 0 & -1 & 0 & 1 & 0 \\
 -4 & -5 & 0 & 0 & 1
 \end{array}
 \left| \begin{array}{c} -1 \\ -1 \\ 0 \end{array} \right.$$

esce di base s_1 ed entra in base x_1

$$\frac{-4}{-1} = 4 \quad \frac{-5}{-\frac{1}{2}} = 10$$

↑
min

$$\bar{R}_1 = -R_1 = 1 \quad \frac{1}{2} \quad -1 \quad 0 \quad 0 \quad 1$$

$$\bar{R}_2 = R_2$$

$$\bar{R}_0 = R_0 + 4\bar{R}_1 = \begin{array}{cccccc} -4 & -5 & 0 & 0 & 1 & 0 \\ 4 & 2 & -4 & 0 & 0 & 4 \\ \hline 0 & -3 & -4 & 0 & 1 & 4 \end{array}$$

$$\left[\begin{array}{ccccc|c} \textcolor{red}{x}_1 & \textcolor{blue}{x}_2 & \textcolor{blue}{s}_1 & \textcolor{red}{s}_2 & \varphi & \\ \hline 1 & \frac{1}{2} & -1 & 0 & 0 & 1 \\ 0 & -1 & 0 & 1 & 0 & -1 \\ \hline 0 & -3 & -4 & 0 & 1 & 4 \end{array} \right]$$

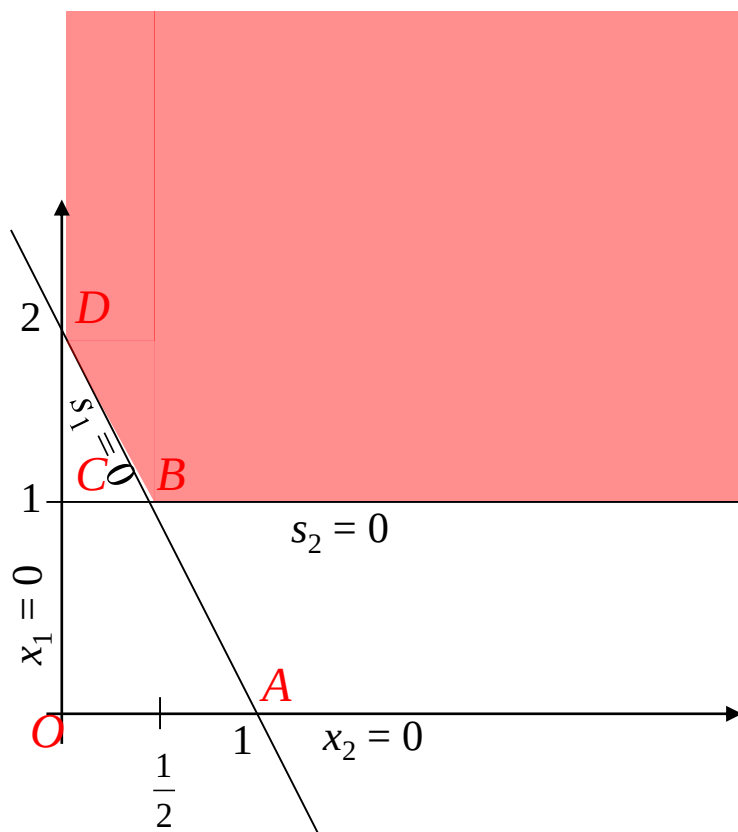
vertice A

Soluzione di base

- **Non Ammissibile**
- **Ottima**

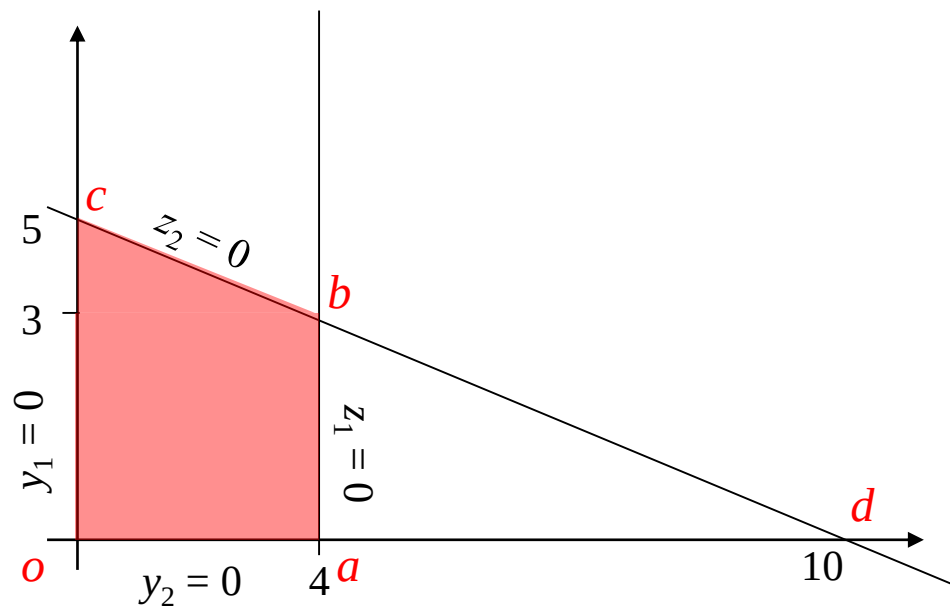
Problema del compratore

(da risolvere)



Problema del produttore

(duale del problema da risolvere)



Vertice A :

$$\begin{array}{c}
 \textcolor{red}{x}_1 \quad \textcolor{blue}{x}_2 \quad \textcolor{blue}{s}_1 \quad \textcolor{red}{s}_2 \quad \varphi \\
 \left[\begin{array}{ccccc|c}
 1 & \frac{1}{2} & -1 & 0 & 0 & 1 \\
 0 & -1 & 0 & 1 & 0 & -1 \\
 0 & -3 & -4 & 0 & 1 & 4
 \end{array} \right]
 \end{array}$$

Soluzione di base

- **Non Ammissibile**
- **Ottima**

Ricaviamo il vertice nel problema duale:

$$\begin{array}{c|cc|c}
 & \textcolor{blue}{x}_2 & \textcolor{blue}{s}_1 & \\
 \hline
 \textcolor{red}{x}_1 & \frac{1}{2} & -1 & 1 \\
 \textcolor{red}{s}_2 & -1 & 0 & -1 \\
 \hline
 \varphi & -3 & -4 & 4
 \end{array}$$

$$\begin{array}{c|cc|c}
 & \textcolor{blue}{z}_1 & \textcolor{blue}{y}_2 & \\
 \hline
 \textcolor{red}{z}_2 & -\frac{1}{2} & 1 & 3 \\
 \textcolor{red}{y}_1 & 1 & 0 & 4 \\
 \hline
 \varphi_D & -1 & 1 & -4
 \end{array}$$

$$\begin{array}{c}
 \textcolor{blue}{z}_1 \quad \textcolor{red}{z}_2 \quad \textcolor{red}{y}_1 \quad \textcolor{blue}{y}_2 \quad \varphi_D \\
 \left[\begin{array}{ccccc|c}
 -\frac{1}{2} & 1 & 0 & 1 & 0 & 3 \\
 1 & 0 & 1 & 0 & 0 & 4 \\
 -1 & 0 & 0 & 1 & 1 & -4
 \end{array} \right]
 \end{array}$$

vertice a: s.b.a. non ottima

vertex a

$$\begin{array}{ccccc|c} z_1 & z_2 & y_1 & y_2 & \varphi_D & \\ \hline -\frac{1}{2} & 1 & 0 & 1 & 0 & 3 \\ 1 & 0 & 1 & 0 & 0 & 4 \\ \hline -1 & 0 & 0 & 1 & 1 & -4 \end{array} \leftarrow \min$$

entra in base y_2 ed esce di base z_2

vertex A

$$\begin{array}{ccccc|c} x_1 & x_2 & s_1 & s_2 & \varphi & \\ \hline 1 & \frac{1}{2} & -1 & 0 & 0 & 1 \\ 0 & -1 & 0 & 1 & 0 & -1 \\ \hline 0 & -3 & -4 & 0 & 1 & 4 \end{array}$$

$\uparrow$
min

esce di base s_2 ed entra in base x_2

$$\bar{R}_2 = -R_2 = 0 \quad 1 \quad 0 \quad -1 \quad 0 \quad 1$$

$$\bar{R}_1 = R_1 - \frac{1}{2}\bar{R}_2 = \begin{array}{cccccc} 1 & \frac{1}{2} & -1 & 0 & 0 & 1 \\ 0 & -\frac{1}{2} & 0 & \frac{1}{2} & 0 & -\frac{1}{2} \\ \hline 1 & 0 & -1 & \frac{1}{2} & 0 & \frac{1}{2} \end{array}$$

$$\bar{R}_0 = R_0 + 3\bar{R}_2 = \begin{array}{cccccc} 0 & -3 & -4 & 0 & 1 & 4 \\ 0 & 3 & 0 & -3 & 0 & 3 \\ \hline 0 & 0 & -4 & -3 & 1 & 7 \end{array}$$

$$\left[\begin{array}{cc|cc|c} \textcolor{red}{x}_1 & \textcolor{red}{x}_2 & \textcolor{blue}{s}_1 & \textcolor{blue}{s}_2 & \varphi & \\ \hline 1 & 0 & -1 & \frac{1}{2} & 0 & \frac{1}{2} \\ 0 & 1 & 0 & -1 & 0 & 1 \\ \hline 0 & 0 & -4 & -3 & 1 & 7 \end{array} \right] \quad \text{vertice } B$$

Soluzione di base

- **Ammissibile**
- **Ottima**

Uso del simplesso duale per determinare la nuova s.b.a.o. dopo che la precedente s.b.a.o. è divenuta non ammissibile a causa della variazione di un termine noto.

$$\begin{array}{ll}
 \min & \varphi = 4x_1 + 5x_2 \\
 \text{s.a} & 3x_1 + x_2 \leq 27 \\
 & 3x_1 + 2x_2 \geq 30 \\
 & x_1 + x_2 = 12 \\
 & x_1, x_2 \geq 0
 \end{array}
 \quad
 \begin{array}{c}
 x_1 \quad x_2 \quad s_1 \quad \bar{s}_2 \quad \varphi \\
 \left[\begin{array}{ccccc|c}
 1 & 0 & .5 & 0 & 0 & 7.5 \\
 0 & 1 & -.5 & 0 & 0 & 4.5 \\
 0 & 0 & .5 & 1 & 0 & 1.5 \\
 \hline
 0 & 0 & -.5 & 0 & 1 & 52.5
 \end{array} \right]
 \end{array}$$

$$\begin{array}{ll}
 \text{Base ottima:} & \mathbf{B} = \begin{array}{c} \begin{matrix} x_1 & x_2 & s_2 \end{matrix} \\ \begin{bmatrix} 3 & 1 & 0 \\ 3 & 2 & -1 \\ 1 & 1 & 0 \end{bmatrix} \end{array}
 \end{array}
 \quad
 \mathbf{B}^{-1} = \frac{1}{2} \begin{bmatrix} 1 & 0 & -1 \\ -1 & 0 & 3 \\ 1 & -2 & 3 \end{bmatrix}$$

Intervallo di b_1 per il quale la base ammissibile ottima rimane ammissibile

$$\underline{\tilde{\mathbf{b}}} = \underline{\mathbf{b}} + \underline{\mathbf{e}}_1 \delta$$

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$$\underline{\tilde{\mathbf{w}}}_B = \mathbf{B}^{-1} \underline{\tilde{\mathbf{b}}} = \mathbf{B}^{-1} \underline{\mathbf{b}} + \mathbf{B}^{-1} \underline{\mathbf{e}}_1 \delta$$

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$$\underline{\tilde{\mathbf{w}}}_B = \mathbf{B}^{-1} \underline{\tilde{\mathbf{b}}} = \mathbf{B}^{-1} \underline{\mathbf{b}} + \mathbf{B}^{-1} \underline{\mathbf{e}}_1 \delta = \begin{bmatrix} \frac{15}{2} \\ \frac{9}{2} \\ \frac{2}{3} \\ \frac{2}{2} \end{bmatrix} + \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{2} \end{bmatrix} \delta = \begin{bmatrix} \frac{15}{2} + \frac{1}{2} \delta \\ \frac{9}{2} - \frac{1}{2} \delta \\ \frac{2}{3} + \frac{1}{2} \delta \\ \frac{2}{2} \end{bmatrix} \quad \begin{cases} \delta \geq -15 \\ \delta \leq 9 \\ \delta \geq -3 \end{cases} \quad -3 \leq \delta \leq 9$$

$$\underline{\tilde{\mathbf{b}}} = \underline{\mathbf{b}} + \underline{\mathbf{e}}_1 \delta$$

$$\underline{\tilde{\mathbf{w}}}_B = \mathbf{B}^{-1} \underline{\tilde{\mathbf{b}}} = \mathbf{B}^{-1} \underline{\mathbf{b}} + \mathbf{B}^{-1} \underline{\mathbf{e}}_1 \delta = \begin{bmatrix} \frac{15}{2} \\ \frac{9}{2} \\ \frac{2}{3} \\ \frac{2}{2} \end{bmatrix} + \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{2} \end{bmatrix} \delta = \begin{bmatrix} \frac{15}{2} + \frac{1}{2} \delta \\ \frac{9}{2} - \frac{1}{2} \delta \\ \frac{2}{3} + \frac{1}{2} \delta \\ \frac{2}{2} \end{bmatrix} \quad \begin{cases} \delta \geq -15 \\ \delta \leq 9 \\ \delta \geq -3 \end{cases} \quad -3 \leq \delta \leq 9$$

$$\tilde{\varphi} = \underline{\mathbf{c}}_B^T \mathbf{B}^{-1} \underline{\tilde{\mathbf{b}}} + \gamma = \underline{\mathbf{c}}_B^T \mathbf{B}^{-1} \underline{\mathbf{b}} + \gamma + \underline{\mathbf{c}}_B^T \mathbf{B}^{-1} \underline{\mathbf{e}}_1 \delta = \frac{105}{2} - \frac{1}{2} \delta$$

Supponiamo si verifichi la variazione $\delta = 10$

$$\delta = 10: \quad \underline{\tilde{\mathbf{w}}}_B = \begin{bmatrix} \frac{25}{2} \\ 2 \\ 1 \\ -\frac{1}{2} \\ 2 \\ \frac{13}{2} \\ 2 \end{bmatrix} \quad \tilde{\varphi} = \frac{95}{2}$$

$$\begin{array}{ccccc|c} x_1 & x_2 & s_1 & \bar{s}_2 & \varphi & \\ \hline 1 & 0 & \frac{1}{2} & 0 & 0 & \frac{25}{2} \\ 0 & 1 & -\frac{1}{2} & 0 & 0 & -\frac{1}{2} \\ 0 & 0 & \frac{1}{2} & 1 & 0 & \frac{13}{2} \\ \hline 0 & 0 & -\frac{1}{2} & 0 & 1 & \frac{95}{2} \end{array}$$

Supponiamo si verifichi la variazione $\delta = 10$

$$\delta = 10: \quad \underline{\tilde{\mathbf{w}}}_B = \begin{bmatrix} \frac{25}{2} \\ 2 \\ 1 \\ -\frac{1}{2} \\ \frac{13}{2} \\ 2 \end{bmatrix} \quad \tilde{\varphi} = \frac{95}{2}$$

$$\begin{array}{ccccc} x_1 & x_2 & s_1 & \bar{s}_2 & \varphi \\ \left[\begin{array}{ccccc|c} 1 & 0 & \frac{1}{2} & 0 & 0 & \frac{25}{2} \\ 0 & 1 & -\frac{1}{2} & 0 & 0 & -\frac{1}{2} \\ 0 & 0 & \frac{1}{2} & 1 & 0 & \frac{13}{2} \\ \hline 0 & 0 & -\frac{1}{2} & 0 & 1 & \frac{95}{2} \end{array} \right] \end{array}$$

$$\begin{array}{ccccc} x_1 & x_2 & s_1 & \bar{s}_2 & \varphi \\ \left[\begin{array}{ccccc|c} 1 & 1 & 0 & 0 & 0 & 12 \\ 0 & -2 & 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 & 6 \\ \hline 0 & -1 & 0 & 0 & 1 & 47 \end{array} \right] \end{array}$$

Teorema degli scarti complementari

Siano

$(\mathbf{x}, \mathbf{s})$ soluzione ammissibile per il problema primale

$(\mathbf{z}, \mathbf{y})$ soluzione ammissibile per il problema duale

Le soluzioni $(\mathbf{x}, \mathbf{s})$ e $(\mathbf{z}, \mathbf{y})$ sono ottime per i rispettivi problemi

se e solo se

$$x_j \cdot z_j = 0 \text{ per ogni } j \quad \text{e} \quad y_i \cdot s_i = 0 \text{ per ogni } i$$

Queste condizioni sono dette *condizioni di complementarità*.

Le condizioni di complementarità si possono anche esprimere nella forma

$$x_j \cdot \left(\sum_{i=1}^m a_{i,j} y_i - p_j \right) = 0 \quad \forall j$$

$$\left(b_i - \sum_{j=1}^n a_{i,j} x_j \right) \cdot y_i = 0 \quad \forall i$$